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SUPERSEDING
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MILITARY SPECIFICATION

RACKS AND CYLINDERS FOR MECHANICAL DISHWASING MACHINES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers racks for handling dishes, compartment type mess trays, service trays, tumblers and flatware in Navy shipboard type and all commercial type mechanical dishwashing machines; and racks and cylinders for dispensing flatware.

* 1.2 Classification. The racks and cylinders shall be of the following types as specified (see 6.1).

Type I - Rack, compartment mess tray/plate, (size compartments).
Type IA - Rack, service tray (15 X 20 inch size), (eight compartments).
Type IB - Rack, compartment mess tray (five compartments).
Type IC - Rack, flatware washing.
Type ID - Rack, tray/plate (16 X 16 inch size).
Type IE - Rack, flatware washing (16 X 16 inch size).
Type IF - Rack, mixed dish (circular shaped).
Type IG - Rack, plate.
Type II - Rack, combination, cup/bowl.
Type IIA - Rack, tumbler.
Type IIB - Rack cup.
Type IIC - Rack cup, compartmented.
Type IID - Rack, cup/bowl (16 X 16 inch size).
Type IIE - Rack, cup/bowl (circular shaped).
Type III - Rack, cylinder, flatware, washing and transporting.
Type IIIA - Rack, cylinder, flatware, washing and transporting (flight type).
Type IIIB - Rack, cylinder, flatware, washing and transporting (skid type with movable handle).
Type V - Cylinder, flatware (plastic).
Type VI - Cylinder, flatware (corrosion-resisting steel).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7320

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.1), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-394 - Plastic Molding Material, Polypropylene, Injection, and Extrusion.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-P-20689 - Plastic, Plastisol (for Coating Metallic Objects).
MIL-M-20693 - Molding Plastic, Polymide (Nylon) Rigid.

STANDARDS

FEDERAL

FED-STD-601 - Rubber, Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

(Copies of specifications, standards, and publications required by suppliers in connection with specific acquisition functions should be obtained as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 1000 16th Street, N.W., Washington, DC 20036.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.
A176 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet, and Strip.
D782 - Method of Test for Shipping Containers in Revolving Hexagonal Drum.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL SANITATION FOUNDATION (NSF)

C-2 - Special Equipment and/or Devices.
Listing of Food Service Equipment.

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, 3475 Plymouth Road, Ann Arbor, MI 48106.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

* 3.1 Description. The racks shall consist of containers for handling dishes, compartment mess- and service-trays, tumblers, flatware, and cylinders. The cylinders are containers for handling flatware.

* 3.2 First article. When specified (see 6.1), the contractor shall furnish a rack or cylinder of the type required for first article inspection and approval (see 4.2.1 and 6.3).

* 3.3 Standard compliance. Racks and cylinders shall meet the requirements of the following sanitary standard: NSF Criteria No. C-2.

* 3.3.1 Certification. As part of the first article test, if a first article is submitted, or prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the racks and cylinders he proposes to furnish under this specification meet the requirements of NSF C-2. Acceptable evidence of meeting the requirements of NSF C-2 shall be the NSF Seal on the finished item and inclusion in the current edition of the NSF Listing of Food Service Equipment.

* 3.3.2 Alternate certification. In lieu of the certification specified in 3.3.1, acceptable evidence of meeting the requirements of the standard will be a certified test report from a nationally recognized testing laboratory, approved by the contracting officer, or acceptable to the medical department of the service for which the racks and cylinders are being acquired attesting that a rack and a cylinder typical in all respects of the units to be supplied under this specification has been examined and tested, and meets all requirements of the applicable standard. When authorized in writing by the

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contracting officer, such examination and tests for alternate certification may be conducted by the manufacturer at the manufacturer's plant when the cognizant inspection authority has determined that adequate facilities and instrumentation are available for conducting the required tests.

* 3.3.3 Applicability. The racks and cylinders shall meet the material, design, operational, and construction requirements, and all other requirements of this specification. The contractor shall select the racks and cylinders to be offered as meeting these requirements from his product lines which satisfy the NSF sanitary standard. Racks and cylinders meeting the operational and design requirements of this specification, but which in other respects do not meet the requirements herein will not be acceptable.

* 3.4 Standard commercial product. The rack or cylinder shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product, shall be included in the rack or cylinder being furnished. A standard commercial product is a product which had been sold or is currently being offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

* 3.5 Interchangeability of items. All racks and cylinders of the same classification furnished under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, and accessories. No deviations will be acceptable without prior written approval of the contracting officer.

3.6 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles, incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specifically specified.

3.6.1 Plastic coating. Plastic coating material for wire base inserts shall conform to the requirements of MIL-P-20689, type I, class 1, except for the following modifications:

(a) The color is to be that which is normally supplied by the manufacture.
(b) When material is submitted to the boiling water test of 4.6.1, swelling shall be not more than 10 percent and shrinkage shall be not more than 2 percent.

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3.6.2 Molding plastic for racks. The plastic material used in making the doby or frame of racks and molded inserts or parts shall be polypropylene. The polypropylene molding material shall meet the requirements of type III, grade A or B, or type I of L-P-394. A certificate of identification will be acceptable.

3.6.3 Molding plastic for cylinders. Plastic molding materials for type V cylinders shall be polyamide (nylon) resin with the necessary fillers, plasticizers and modifiers or polypropylene (see 3.6.2). The nylon molding materials shall meet the requirements of composition A, polyhexamethylene-amide, type I, general purpose, of MIL-M-20693. A certificate of identification will be acceptable.

* 3.6.4 Corrosion-resisting steel. Corrosion-resisting steel shall conform to ASTM A167 or A176, as specified (see 6.1).

3.6.5 Steel wire. Steel wire used in construction of inserts shall conform to any one of AISI type C1005 to C1008.

3.6.6 Plated steel. When specified (see 6.1), this material shall be of the quality used on standard commercial racks and cylinders and shall meet the sanitary standard as specified herein.

* 3.7 Design. The bottom of racks and cylinders shall be designed to prevent the protrusion of applicable ware through the openings. Empty racks shall be suitable for stacking.

3.7.1 Dishwashing racks.

* 3.7.1.1 Type I and II racks. Type I and II racks shall be suitable for operation in commercial type dishwashing machines having a track width of 20-1/8 inches, and Navy shipboard type dishwashing machines with center-to-center indexing lug spacing approximately 19-1/2 inches and track width of 20-7/16 inches. Fouling and respacing of tracks and indexing lugs on either type of machine shall not be permitted. Racks are not limited in length to fit between the 19-1/2 inch lug spacing; however, rack indexing shall be compatible with the 19-1/2 inch lug spacing. Racks shall be designed for double- or four way-indexing, as specified (see 6.1). When specified (see 6.1), type II rack shall be designed so that the bottom surface of cups, bowls, and tumblers when placed in the rack, is tilted at an angle not less than 10° from the horizontal to facilitate drainage.

* 3.7.1.2 Type IB, IC, IG, IIA, IIB, and IIC rack. Racks shall be suitable for operation in commercial type dishwashing machines having a track width of 20-1/8 inches without fouling, and without respacing of tracks or indexing lugs. The racks shall be approximately 19-34 inches wide by 19-3/4 inches long. Racks shall be designed for double- or four way-indexing as specified (see 6.1). When specified (see 6.1), types IIA, IIB, and IIC racks shall be designed so that the bottom surface of the cups and tumblers when placed in the rack, is tilted at an angle not less than 10° from the horizontal to facilitate drainage.

* 3.7.1.3 Type ID, IE, and IID rack. Type ID, IE, and IID racks shall be 16 inches square, and shall be suitable for use in single tank commercial type dishwashing machines having a width not less than 16-1/8 inches and not more

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than 16-1/4 inches. When specified (see 6.1), type IID rack shall be designed so that the bottom surface of the cups and bowls when placed in the rack, is tilted at an angle not less than 10° from the horizontal to facilitate drainage.

* 3.7.1.4 Type IF and IIE racks. Type IF and IIE racks shall be suitable for use in a single tank dishwashing machine having a round or circular configuration. When specified (see 6.1), the racks shall be designed so that the bottom surface of the mixed dishes, cups, and bowls when placed in the rack, is tilted at an angle not less than 10° from the horizontal to facilitate drainage.

3.7.1.5 Type IA rack. Type IA rack shall be suitable for operation in commercial type dishwashing machines having a track width of 20-1/8 inches without fouling and without respacing of tracks or indexing lugs. The rack shall be approximately 19-3/4 inches, except that one end shall be cut or opened to accommodate the 20-inch dimension of 15 by 20 inch trays.

* 3.7.1.6 Type IG rack. Type IG rack shall be designed to support general purpose plates and saucers in near vertical position for use in a dishwashing machine. Unless otherwise specified (see 6.1), indexing shall be in accordance with the manufacturer's standard practice.

* 3.7.2 Flatware racks and cylinders. Type III, IIIA, and IIIB racks shall be the basket-type for transporting six type V or VI cylinders with contents to, through, and from dishwashing machines. Type III shall be for use in conjunction with rack type dishwashers. Types IIIA and IIIB, provided with runners or supports shall be for use with type IIIB rack shall have a slotted swivel handle. The racks shall be designed for double- or four way-indexing, as specified (see 6.1). Type V and VI cylinders shall be suitable for washing and stowing flatware when used in conjunction with type III, IIIA, and IIIB racks.

* 3.8 Construction.

3.8.1 Dishwashing racks.

* 3.8.1 Type I, IA, IB, IC, IG, II, IIA, IIB, and IIC rack. The rack shall be constructed to meet design requirements of 3.7.1. A frame or body shall be constructed of plastic specified in 3.6.2. The rack may be of integral construction having frame or doby and division structure molded together as a unit, or constructed with molded plastic frame and separate insert. When wire insert is provided, the coating material shall be in accordance with 3.6.1. The coating shall be applied as specified in 3.8.1.3. All exposed (uncoated) metal, such as clamps for securing inserts to frames, shall be corrosion-resisting steel or plated steel as specified in 3.6.6. The diagonal distance between corners of finished rectangular racks shall vary not more than ±1/8 inch. Type II, IIA, IIB, and IIC loaded racks shall be capable of stacking. Ware in stacked racks shall not be in contact with the rack positioned above.

* 3.8.1.2 Type ID, IE, IF, IID, and IIE rack. The rack shall be constructed to meet design requirements of 3.7.1. A frame or body shall be constructed of steel wire specified in 3.6.4 and 3.6.5, or plastic specified in 3.6.2.

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Plastic racks may be of integral construction having frame of body and division structure molded together as a unit, or constructed with molded plastic frame and separate insert. When wire insert is provided, the coating material shall be in accordance with 3.6.1. The coating shall be applied as specified in 3.8.1.3. Wire frame top and bottom main members shall be constructed of figure 8 section not less than 0.318 inch by 0.15 inch. Other frame members and stays shall have a minimum diameter of 0.177 inch. The diagonal distance between corners of finished rectangular racks shall vary not more than ±1/8 inch. The diameter of round or circular racks shall vary not more than ±1/8 inch.

* 3.8.1.3 Coated wire inserts. Wire inserts shall consist of minimum 12-gage wire coated with plastic material specified in 3.6.1 in accordance with the following:

(a) The wire insert shall be thoroughly cleaned, dipped in a suitable primer that will assure adhesion of the plastic to the metal, and then baked until the wire has a uniform temperature throughout of 350° Fahrenheit (F).
(b) The primed insert, immediately after removal from the oven, shall be immersed in the vinyl dispersion specified in 3.6.1 until a coating has cured on all primed wire surface. Combined thickness of plastic and primer shall average 0.032 inch with reasonably uniform distribution.
(c) After coating, the insert shall be baked until complete polymerization has taken place.

3.8.1.3.1 Finished coating. The finished plastic coating shall be reasonably smooth and free from excessive lumps, drips, air pockets or holes, and shall be uniformly and securely bonded to the wire. The finished insert shall be capable of withstanding tests specified in 4.6.1 without change in appearance, weight, and condition.

* 3.8.1.4 Rack sides. Rack sides shall be provided with openings or shall be solid, as specified (see 6.1). Solid sides shall be impervious to wash- and rinse-water.

3.8.1.5 Type I rack. The rack shall be suitable for holding six compartmented mess trays having approximate dimensions of 15-1/2 inches long by 11-5/8 inches wide by 3/4-inch deep, or twelve 9-inch dinner plates. The rack shall be divided to hold six trays with a minimum clear space of 2-5/8 inches between trays. When placed in the rack for washing, the trays shall be at an angle of approximately 18° from the vertical.

3.8.1.6 Type IA rack. Rack shall be suitable for holding not less than 7 15- by 20-inch cafeteria type service trays. The rack shall be divided to hold the trays in a suitable position for efficient cleaning and sanitizing.

3.8.1.7 Type IB rack. Rack shall be suitable for holding not less than 5 compartmented mess trays having approximate dimensions of 15-1/2 inches long by 11-5/8 inches wide by 3/4-inch deep. Clear space between each tray shall be not less than 3 inches. Trays, when placed in the rack for washing, shall incline or slant approximately 30° from the vertical.

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3.8.1.8 Type IC rack. Rack shall be suitable for laying silverware flat for washing and sanitizing. Bottom of rack shall be of a design to permit free flow of water and prevent loss of silverware, or bottom of rack shall be fitted with a flat insert. Rack shall be of a design to prevent overloading with silverware.

3.8.1.9 Type ID rack. Rack shall be divided into seven equal compartments for holding not less than seven trays or seven dinner plates. Design shall be such that items inserted in the rack will be properly positioned for efficient cleaning and sanitizing in a single tank dishwashing machine.

3.8.1.10 Type IE rack. Rack shall be suitable for laying silverware flat for washing and sanitizing in a single tank dishwashing machine. Bottom of rack shall be of a design to prevent silverware from protruding, or bottom of rack shall be fitted with a flat insert, and shall permit a free flow of water. Rack shall be of a design to prevent overloading of silverware.

3.8.1.11 Type IF rack. Rack shall be suitable for washing and sanitizing plates in a single tank dishwashing machine of circular shape. The rack shall have a capacity of eight dinner plates or fifteen dessert plates.

* 3.8.1.12 Type IG rack. Rack shall be suitable for holding not less than sixteen 9-inch diameter plates of 24 oval plates (5-3/4 by 4-1/2 inches) in a satisfactory position for proper washing and sanitizing in a dishwashing machine.

* 3.8.1.13 Type II rack. This rack shall be suitable for holding cups and bowls in the inverted position. Rack shall hold not less than sixteen 10-1/2 ounce cups or nine 5-3/4 inch diameter bowls. The frame or body of the rack should have an inside height not less than 3 inches.

* 3.8.1.14 Type IIA rack. Rack shall be divided into not less than 36 compartments suitable for washing and stowing tumblers having maximum dimensions of 4 inches high by 2-3/4 inches in diameter. Inside height of rack shall be not less than 4 inches of more than 5-1/8 inches.

3.8.1.15 Type IIB rack. Rack shall hold not less than eighteen 10-1/2 ounce cups. Provision shall be designed into the rack for individual spacing or positioning of the cups.

3.8.1.16 Type IIC rack. Rack shall be divided into 16, 24, or 25 compartments, as specified (see 6.1), for washing and stowing cups or mugs. Separation between compartments shall be such as to prevent contact between individual cups or mugs.

3.8.1.17 Type IID rack. The rack shall be suitable for holding cups and bowls in the inverted position. Rack shall hold not less than fifteen 10-1/2 ounce cups or five 5-3/4 inch diameter bowls.

3.8.1.18 Type IIE rack. Rack shall be suitable for washing and sanitizing cups, bowls, and silverware in a single tank dishwashing machine of circular shape. The rack shall have a capacity for fourteen 10-1/2 ounce cups or seven 5-3/4 inch diameter bowls. Bottom design shall be suitable for washing silverware.

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* 3.8.2 Flatware cylinder washing and transporting rack. The flatware cylinder washing and transporting rack shall be type III, IIIA, and IIIB. The rack shall consist essentially of a basket type body and a handle. The body shall have rounded corners with a radius of 1/2 to 3/4 inch. The rack shall hold six fully loaded type V or VI cylinders supported by the cylinder flanges when the rack is lifted and carried by the handle. There shall be no distortion of the rack. Each rack shall be suitable for stacking when empty.

* 3.8.2.1 Type III rack. The type III rack shall be constructed of corrosion-resisting steel or plastic. The body shall have overall dimensions of approximately 10-11 inches in width and 14-15 inches in length. The height exclusive of the handle, shall be not less than 20 1/4 and not more than 4-1/4 inches. The handle shall have its axis parallel to the length of the rack, and shall be securely fastened to the basket at the ends. The grip of the handle shall be of a height to clear by 1 inch the protruding end of an 8-1/2 inch long knife resting in any position in any cylinder. The handle shall prevent the rack from tipping to either side due to unbalanced loading of the cylinders when the handle is fully extended in the carrying position.

* 3.8.2.2 Type IIIA rack. The type IIIA rack shall have a corrosion-resisting steel body, tubular skid type stand or support, and handle. The approximate overall dimensions shall be 10 inches wide by 15 inches long by 17-7/8 inches high to top of the handle. The tubular stand or support shall durably support the rack and loaded cylinders in a flight type dishwasher to prevent contact of cylinder bottoms with conveyor pins or pegs. The handle shall be secured to the rack body with locktight cap nuts.

* 3.8.2.3 Type IIIB rack. The type IIIB rack shall be constructed similar to type IIIA, except the handle shall be a slotted swivel type as required for type III racks.

3.8.3 Type V flatware cylinders. Type V cylinders shall be molded of plastic material conforming to 3.6.2 or 3.6.3. The overall height of the cylinders shall be 5-1/2 inches ±3/32 inch. Top of the cylinders shall have a molded flange extending outward approximately 1/4 inch. Thickness of flange shall be not less than thickness of body. The outside diameter (od) of cylinders at the top, directly under the flange, shall be 3-13/16 inches ±1/16 inch. The body of the cylinder may be tapered to an od of not less than 3-9/16 inches at the bottom. Cylinder wall and bottom shall be perforated or slotted to permit free flow of water from the cylinder. Perforations or slots shall be of a size to prohibit flatware from falling out of the cylinder. All holes or slots shall be so located as not to adversely affect the durability of the cylinder. Wall and bottom thickness, exclusive of reinforcement ribs, shall be not less than 0.105 inch.

3.8.4 Type VI flatware cylinders. Type VI cylinders shall be made of corrosion-resisting steel and have a finished thickness not less than 0.0375 inch and not more than 0.0437 inch. Overall height of the cylinders shall be 5-1/2 inches ±3/32 inch. Top of the cylinders shall be flanged out from the body at an approximate 90° angle, not less than 1/4 inch and not more than 3/8 inch. Outside diameter, directly under flange, shall be 3-3/4 inches ±1/16 inch. Body of cylinders may be tapered to an od not less than 3-9/16 inches at the bottom. Cylinder wall and bottom shall be perforated to permit free flow of water from the cylinder. Perforations shall be of a size to prohibit flatware from falling out of the cylinder. All holes shall be so located as not to adversely affect the durability of the cylinder.

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3.9 Workmanship.

* 3.9.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

* 3.9.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

* 3.9.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

* 3.9.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

* 3.9.5 Moldings. All moldings shall be sound and free from patching, warping, or any other defect which reduces the moldings ability to perform its intended function.

* 3.10 Marking identification. Identification marking shall be permanently and legibly marked on the rack and cylinder, or on a corrosion-resisting metal plate securely attached to the rack or cylinder and shall include model, serial number, name, and trademark.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Standard compliance. Proof of compliance with the standards cited in 3.3 shall be made available to the contracting officer or his authorized representative. Certification specified number 3.3.1 or 3.3.2 will be accepted as evidence of compliance.

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* 4.2 Classification of inspection. The inspection requirements specified herein shall be classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

* 4.2.1 First article inspection. First article inspection shall be performed on one complete rack or cylinder when a first article sample is required (see 3.2, 6.1, and 6.3). This inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be a standard production item from the contractor's current inventory, provided the rack or cylinder meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining racks and cylinders to be furnished under the contract.

* 4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on sample racks and cylinders selected in accordance with 4.4. This inspection shall include the examination of 4.5, the tests of 4.6, the materials inspection of 4.7, and the packaging inspection of 4.8.

* 4.3 Inspection lot. All units of the same type and style manufactured under the same conditions and offered to the Government for delivery at the same time, shall be considered a lot for purpose of inspection.

* 4.4 Sampling. A random sample shall be selected from each lot in accordance with MIL-STD-105. The sample unit shall be one completely fabricated rack or cylinder of the same type, as applicable.

* 4.4.1 Sampling for examination. The inspection level shall be level II with an Acceptable Quality Level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

* 4.4.2 Sampling for tests. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

* 4.5 Examination.

* 4.5.1 Process examination. Examination shall be made of the following operations to establish that no deviation is made from the indicated requirements. Whenever a deviation is noted, the contractor shall take corrective action. Failure of the contractor to take corrective action shall be cause for rejection of affected end-item lots. Examinations shall be as follows:

(a) Workmanship (see 3.9).
(b) Baking of primer, where applicable (see 3.8.1.3).
(c) Dipping into plastic immediately after removal from oven and baked, where applicable (see 3.8.1.3).

* 4.5.2 Visual examination. Racks and cylinders shall be examined visually in accordance with the classification of defects set forth in table I.

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TABLE I. Classification of visible defects.

Categories	Defects
Major	
	Plastic coating (see 3.6.1):
101	Not as specified.
102	Not uniform and securely bonded to metal.
103	Not coated where required.
104	Corrosion-resisting steel not equivalent to ASTM specifications.
105	Plastic coating blistered, peeled, or containing cavities.
	Molded plastic (see 3.6.2 and 3.6.3):
106	Not uniform, sound, or smooth.
107	Not free from flashings, cracks, blowholes, or airholes.
	Metallic plating (see 3.6.4 and 3.6.6):
108	Type of coating not as specified.
109	Not finished or coated where required.
110	Coating blistered, peeled, chipped, or area of thin or incomplete coating.
	Design (see 3.7):
111	Characteristics not in accordance with specified requirements.
112	Part missing or not specified type, size, or material.
	Construction and workmanship (see 3.9):
113	Fractured, split, punctured, dented, deteriorated, bowed, or malformed, or otherwise impaired.
114	Misplaced, loose, or not in proper alinement.
115	Sharp burr, sliver, or splinter that may be injurious to personnel.
116	Not connected or joined as specified or poorly accomplished.
	Welding and brazing (see 3.9.4):
117	Not type required, missing, incomplete, burnholes, cracked, fractured, or otherwise not fused.
118	Slag inclusion, slight undercut, not smooth and uniform scale or flux deposits not removed.
	Assembly:
119	Unit perceptibility out of square or alinement.
Minor	
	Marking for identification (see 3.10):
201	Missing, incomplete, not legible.

* 4.5.3 Dimensional examination. Examination shall be made of the end item to determine compliance with dimensional requirements. Any dimension that exceeds the specific tolerance shall be classified a defect.

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4.6 Tests.

4.6.1 Immersion tests for plastic-coated inserts. One plastic-coated insert, representing the manufacturer's item to be furnished under the contract, shall be tested to determine conformance to the requirements specified in 3.8.1.3.1. The tests shall consist of the following:

(a) Swelling and shrinkage tests on plastic coating material shall be conducted in boiling water for 5 hours. These tests shall be conducted in accordance with Method 6211 of FED-STD-601, with the time and temperature exceptions noted.
(b) Immersion for 10 minutes in the following solution at room temperature:

(1) Concentrated hydrochloric acid.
(2) 50 percent sulfuric acid.
(3) 35 percent sodium hydroxide.

4.6.2 Cylinders. Cylinders shall be tested to determine required resistance under 3.6.2 or 3.6.3 as applicable.

* 4.6.3 Strength test for racks, types III, IIIA, and IIIB. Fill six cylinders, type V or VI to their capacity with flatware (knives, forks, and spoons). For racks, types III, IIIA, and IIIB, place the filled cylinders in the rack to be tested and suspend the rack from the center of the rack handle. After a period of not less than 5 minutes, remove the cylinders and examine the rack and handle for distortion.

* 4.6.4 Impact test for types I to IG and II to IIE racks. One complete rack of type I to IG and II to IIE racks only, shall be supplied under the contract for this test. Racks shall be subjected to not less than 500 impacts in a 14-foot hexagonal revolving drum constructed in accordance with ASTM D782. Racks after impact testing should not show damage, impairment, or prohibit use of the racks for the purpose intended.

* 4.6.5 Bow test for all types of racks. One complete rack of each type shall be supplied under the contract for this test. One complete rack of each type furnished shall be tested to determine resistance against bowing when subjected to environmental conditions similar to those anticipated to be encountered during service use. Racks shall be uniformly loaded with 19 pounds deadweight, immersed and suspended in a tank of 180°F water for a period of 3 minutes. Upon removal the rack shall be permitted to cool to ambient room temperature. The bottom surface of the cooled rack shall be inspected to insure a flat and uneven surface. Racks after completion of testing for bowing shall show no evidence of permanent deformation.

4.7 Materials inspection. The contractor shall certify in writing that the materials used in the construction of the racks and cylinders are in accordance with the requirements of 3.6.1 and 3.6.6. Such certification shall document actual tests, examinations, or other verifiable quality data. In addition, the contractor shall submit evidence that the plastic material used in the construction of the racks, inserts, or parts, and cylinders is polypropylene or polyamide resin in accordance with 3.6.2 and 3.6.3, as applicable. Copies of the contractor's invoices or purchase orders will be acceptable. Copies of test reports shall be furnished that indicate the material has been tested in accordance with the applicable specification.

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* 4.8 Packaging inspection. The preservation, packing, and marking of the racks and cylinders shall be inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.1).

5.1.1 Level A. Each rack, or not more than six cylinders, shall be preserved method III in accordance with MIL-P-116 using fiberboard boxes conforming to PPP-B-636, class weather-resistant.

5.1.2 Commercial. Racks, or cylinders, shall be packaged in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.1).

5.2.1 Levels A and B. Packing shall be in accordance with MIL-STD-794. Containers shall be selected from table II for the appropriate level.

5.2.2 Commercial. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

* 6.1 Ordering data. Acquisition documents sould specify the following:

(a) Title, number, and date of this specification.
(b) Type of racks and cylinders required (see 1.2).
(c) When specifications and standards shall be other than as specified (see 2.1.1).
(d) When first article sample is required (see 3.2, 4.2.1, and 6.3).
(e) Type of steel required (see 3.6.4 and 3.6.6).
(f) When double- or four way-indexing of racks is required (see 3.7.1.1, 3.7.1.2, and 3.7.2).
(g) When racks shall be designed so that the bottom surface of ware is tilted not less than 10° from the horizontal to facilitate drainage (see 3.7.1.1, 3.7.1.2, 3.7.1.3, and 3.7.1.4).
(h) When indexing of type IG racks shall be different (see 3.7.1.6).
(i) Whether rack sides shall have openings or shall be solid (see 3.8.1.4).
(j) Number of compartments required in the type IIC rack (see 3.8.1.16).
(k) Level of preservation and level of packing (see 5.1 and 5.2).

* 6.2 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

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* 6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete rack or cylinder, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examinations, tests, and approval of the first article.

* 6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-R-24039C RACKS AND CYLINDERS FOR MECHANICAL DISHWASHING MACHINES

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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